



HR Analytics: Job Change of Data Scientists

Predicting Candidate Job-Change Likelihood

1. The Problem

A company offers free data-science training. Some trainees stay; others use it as a stepping stone.

Training every candidate is costly. The company needs to identify — in advance — who is likely to leave, so retention and hiring efforts focus on the right people.

Framed as binary classification:

Target = 1 → Looking for a new job

Target = 0 → Not looking

2. Dataset Overview

Source: HR Analytics — Job Change of Data Scientists (Kaggle)

19,158

Training Rows

14

Columns (13 features + target)

2,129

Test Rows (unlabeled)

75.1% / 24.9%

Class Balance (imbalanced)

Key Data Quality Issues & Handling:

- Missing values up to 32% — kept as own category (carries signal)
- Disguised numeric text ('<1', '>20', 'never') — mapped to integers
- High-cardinality city (123 values) — dropped, used city_development_index
- Engineered feature: missing company info flag → 2× higher job-hunt rate

3. Methodology & Approach

1

Preprocessing

Cleaned, encoded, stratified 80/20 split

2

5 Models

Logistic Regression, KNN, SVM, Random Forest, XGBoost

3

Class Imbalance

class_weight='balanced' / scale_pos_weight applied

4

Threshold Tuning

19 thresholds (0.05–0.95), F1-maximizing chosen per model

5

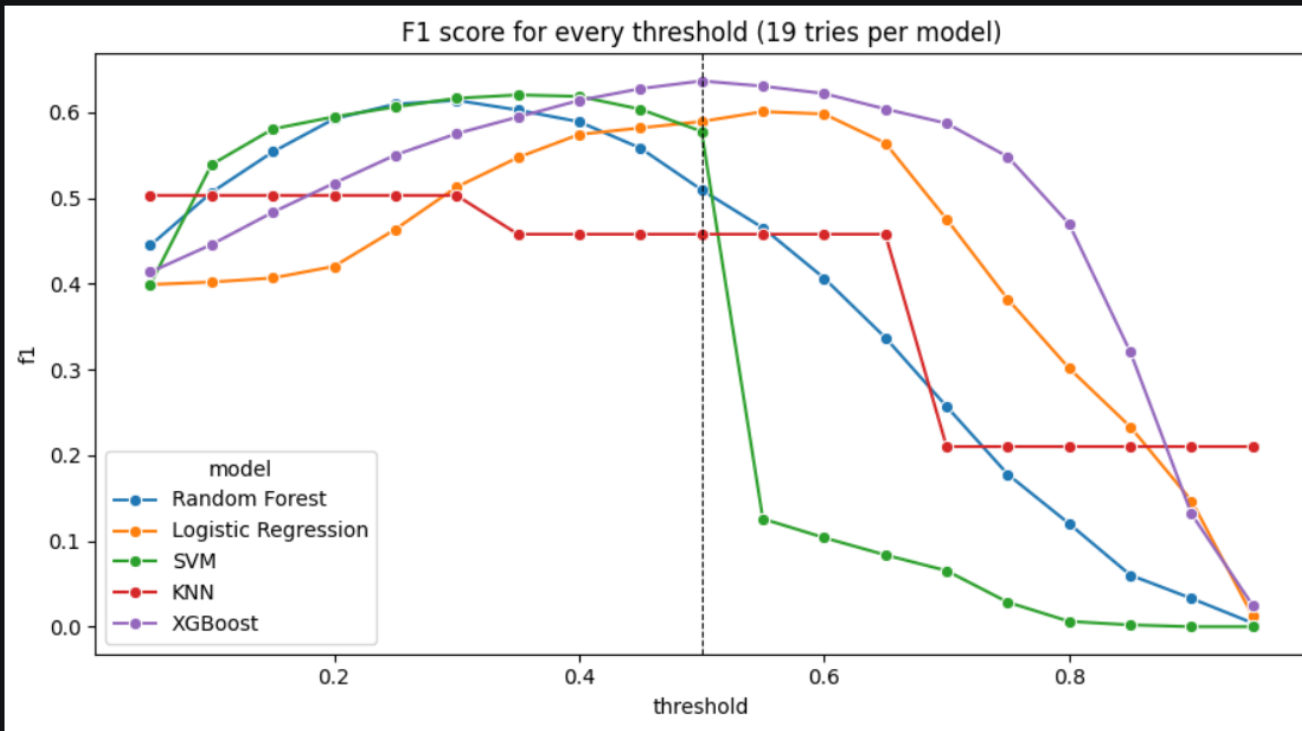
Evaluation

F1-score as primary metric (balances precision & recall)

4. Results — Model Comparison

Best result of each model over 19 thresholds:

model	threshold	f1	accuracy
XGBoost	0.50	0.6364	0.7868
SVM	0.35	0.6201	0.7672
Random Forest	0.30	0.6138	0.7714
Logistic Regression	0.55	0.6006	0.7539
KNN	0.05	0.5029	0.6203



Best F1 per Model

XGBoost: F1 = 0.6364 (thresh 0.50)

SVM: F1 = 0.6201 (thresh 0.35)

Random Forest: F1 = 0.6138 (thresh 0.30)

Logistic Regression: F1 = 0.6006 (thresh 0.55)

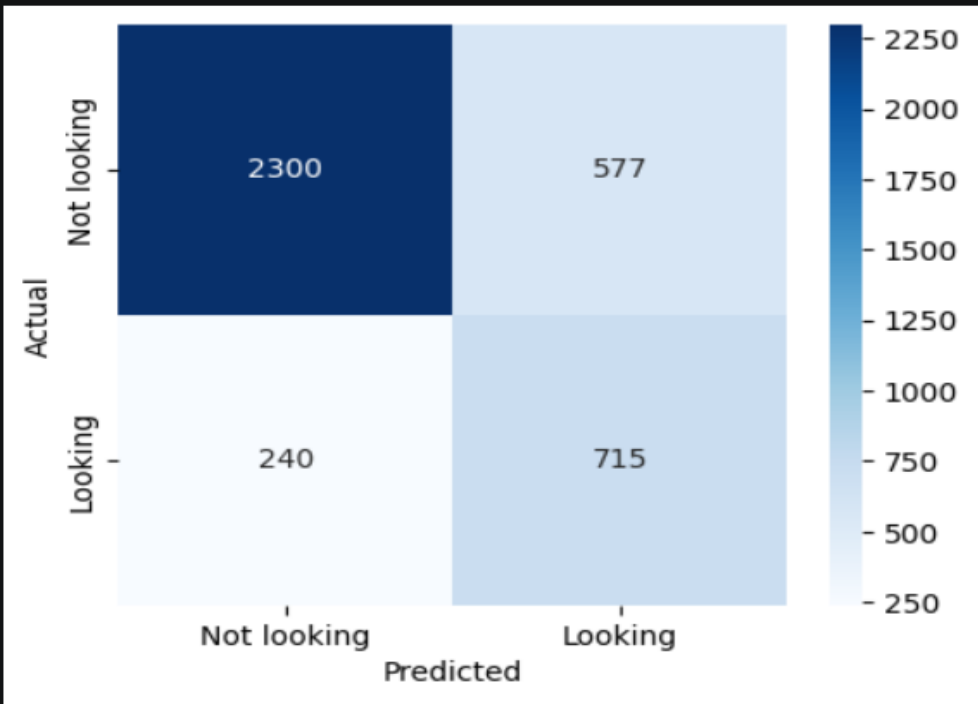
KNN: F1 = 0.5029 (thresh 0.05)

4. Results — Final Model: XGBoost

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Final choice: XGBoost | threshold = 0.5 | F1 = 0.6364
Classification report:
      precision    recall  f1-score   support

     0       0.91      0.80      0.85     2877
     1       0.55      0.75      0.64      955

 accuracy      0.79      3832
 macro avg     0.73      0.77      0.74      3832
 weighted avg   0.82      0.79      0.80      3832
```



Performance Metrics

- Precision (class 1): 0.55
- Recall (class 1): 0.75
- F1-Score (class 1): 0.64
- Overall Accuracy: 0.79

5-Fold Cross-Validation

Mean F1 = 0.613 ± 0.005 (stable across folds)

5. Business Impact & Candidate Ranking

HR teams need a prioritized shortlist, not a flat flag list.

Precision@k (Validation Set)

k = 10: Precision = 0.60 (6 / 10 correct)
k = 20: Precision = 0.60 (12 / 20 correct)
k = 50: Precision = 0.66 (33 / 50 correct)
k = 100: Precision = 0.61 (61 / 100 correct)

Business Value

- ✓ 2.4–2.6× better than random guessing
- ✓ Focus retention budget on high-risk candidates
- ✓ Reduce wasted outreach to stable trainees
- ✓ Actionable Top-10 list for limited HR capacity

Key Insight: Missing company info is a strong predictor — candidates without it are 2× more likely to job-hunt.



Thank You
